RF_CHAIN_1

File: chain1.kicad_sch

RF_CHAIN_2

File: chain2.kicad_sch

RF_CHAIN_3

File: chain3.kicad_sch

RF_CHAIN_4

File: chain4.kicad_sch

LO_SYNTH

File: lo.kicad_sch

CAL_AND_TEST_IO

File: cal.kicad_sch

POWER_SEQUENCING

File: power.kicad_sch

MCU_SAFETY

File: mcu.kicad_sch

PWR_MAIN

File: pwr_main.kicad_sch

PWR_LO

File: pwr_lo.kicad_sch

PWR_NEG

File: pwr_neg.kicad_sch

PA_INTERLOCK

File: interlock.kicad_sch

Sheet: /
File: openru_x6_tile.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 1/25

RF chain 1 (topology A): N310 ch0 IF 1.8 GHz <-> LTC5553 <-> 8.4 GHz <-> cal coupler <-> ADAR1000 UBF1 <-> 4x ADTR1107 <-> 4 feeds.

BF_1

File: bf1.kicad_sch

FE_1

File: fe1.kicad_sch

CONV_1

File: conv1.kicad_sch

J101

C1_IF_N310

1

SMA IF1 (N310 ch0)

GND

SMA footprint (edge-launch vs vertical) UNRESOLVED.

Sheet: /RF_CHAIN_1/
File: chain1.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

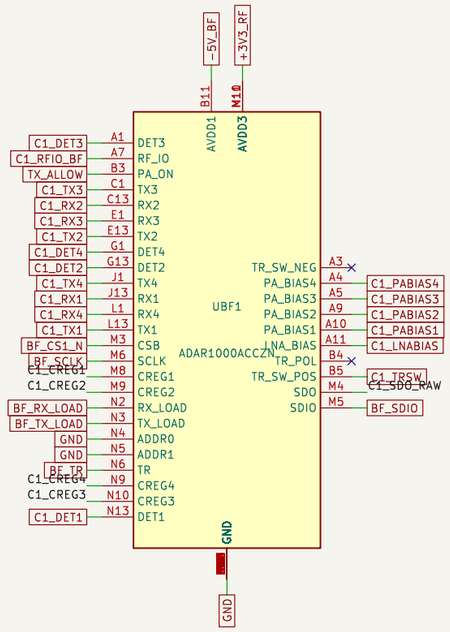
KiCad E.D.A. 9.0.9 | Id: 2/25

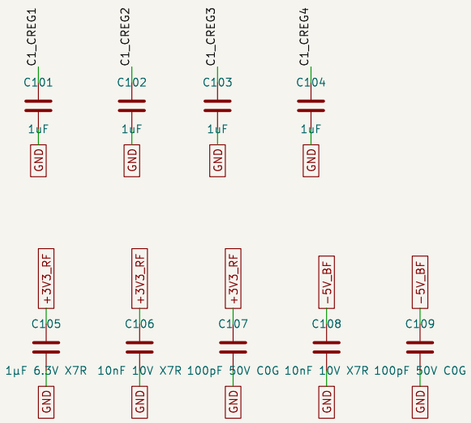
UNRESOLVED: AVDD3 / AVDD1 bypass capacitor values are not given in the ADAR1000 datasheet -> take from ADAR1000-EVALZ schematic.

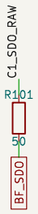
PA_BIASx = 0 V at power-up (DS p41): PA drain stays OFF until firmware sets PA_BIAS OFF = -1.75 V (ADTR1107 VGG_PA abs max -2 V).

SDO errata (DS p47-49): non-addressed SDO not tristated -> read back one chip at a time (separate CSB).

H2-B GATE-BIAS LIMIT (safety): EXT_PAx_BIAS_OFF/ON and EXT_LNA_BIAS_OFF/ON must stay <= 0x5D (~-1.75 V nominal, V = -code x 4.8/255). NEVER use ADAR1000 Rev B p45 values 0x85 (~-2.5 V) or 0x68 (~-1.96 V): ADTR1107 Rev C abs max VGG_PA -2..0 V, VGG_LNA -2..+0.2 V. Measure PA_BIASx with VDD_PA OFF before enabling the drain.







decoupling per ADI Stingray 02_059522 Rev B (per-tile set); per-pin values not documented

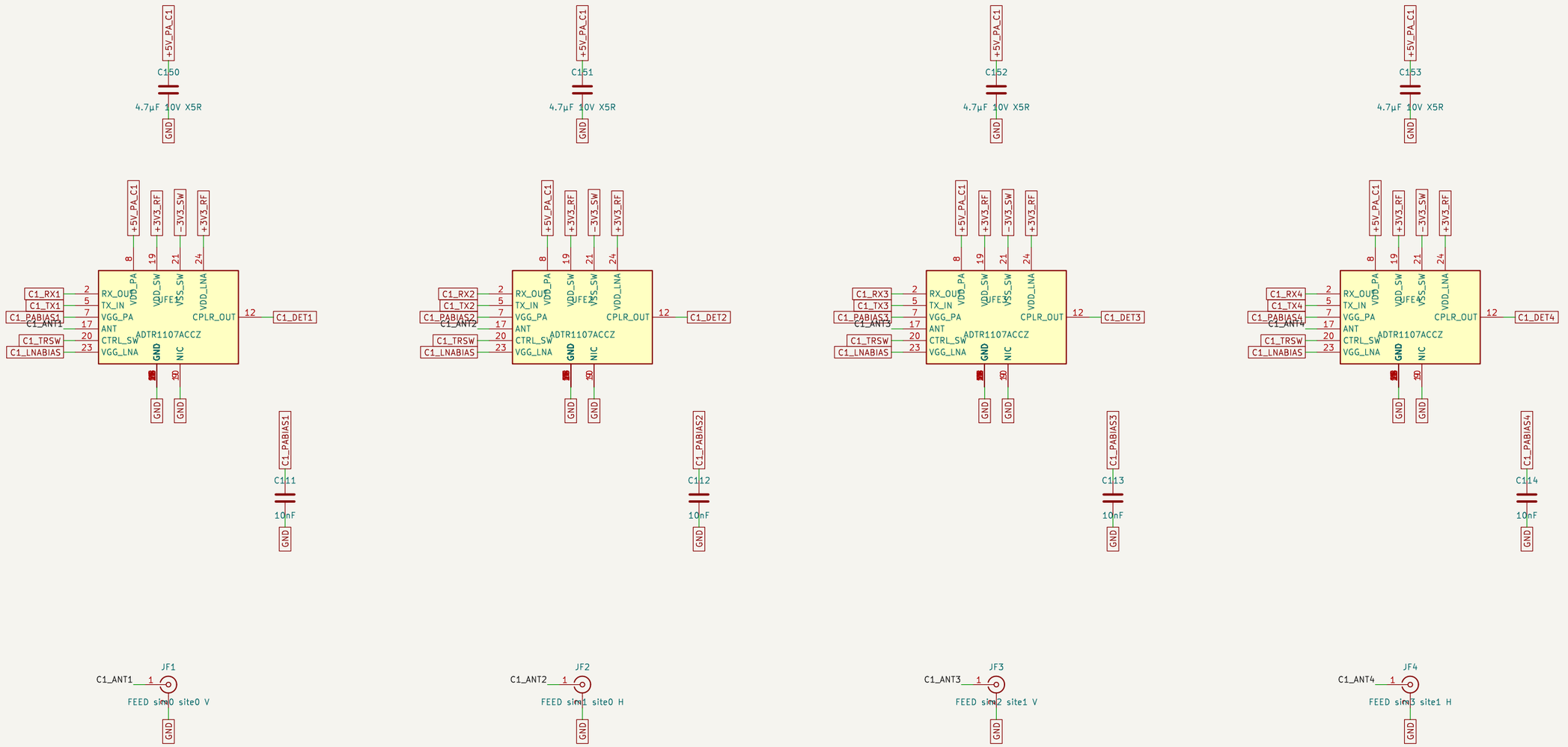
Sheet: /RF_CHAIN_1/BF_1/
File: bf1.kicad_sch

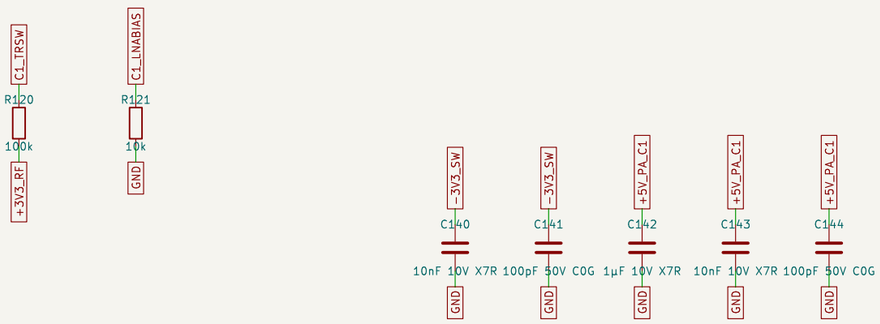
Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.9 | Id: 3/25

UNRESOLVED: VDD_PA / VDD_LNA / VDD_SW / VSS_SW bypass values not in ADTR1107 Rev C (Fig 89 shows only 10 nF on VGG_PA).

Feed ownership = H1_FEED_POLARIZATION_MAP.csv (topology A). LNA kept powered in TX and pinched via LNA_BIAS (Stingray style) — confirm with ADI (ADTR1107 TX sequence lists VDD_LNA = 0 V).





decoupling per ADI Stingray 02_059522 Rev B (per-tile set); per-pin values not documented

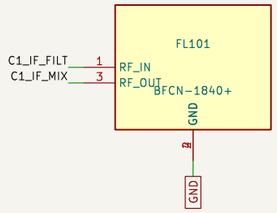
Sheet: /RF_CHAIN_1/FE_1/
File: fe1.kicad_sch

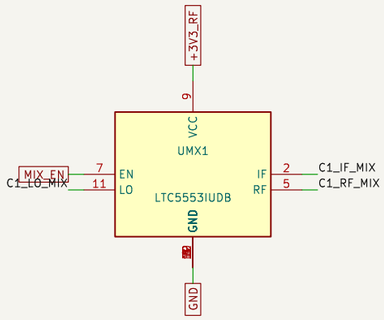
Title:

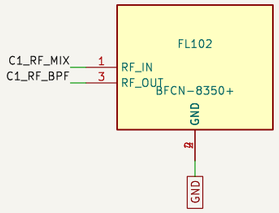
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.9 | Id: 4/25

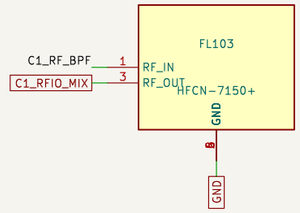
IF 1.75–1.85 GHz: N310 <–> DC block <–> BFCN–1840+ <–> LTC5553 IF (single–ended 50 ohm, no balun).

RF: LTC5553 RF <–> BFCN–8350+ <–> HFCN–7150+ <–> cal coupler (cal sheet) <–> ADAR1000 RF_IO. No DC blocks: RF_IO and LTC5553 RF are both DC–grounded.

















Sheet: /RF_CHAIN_1/CONV_1/
File: conv1.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.9 | Id: 5/25

RF chain 2 (topology A): N310 ch1 IF 1.8 GHz <-> LTC5553 <-> 8.4 GHz <-> cal coupler <-> ADAR1000 UBF2 <-> 4x ADTR1107 <-> 4 feeds.

BF_2

File: bf2.kicad_sch

FE_2

File: fe2.kicad_sch

CONV_2

File: conv2.kicad_sch

J201

C2_IF_N310

1

SMA IF2 (N310 ch1)

GND

SMA footprint (edge-launch vs vertical) UNRESOLVED.

Sheet: /RF_CHAIN_2/
File: chain2.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

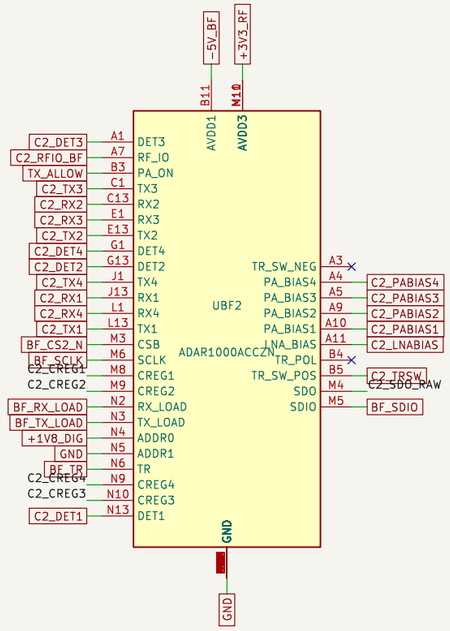
KiCad E.D.A. 9.0.9 | Id: 6/25

UNRESOLVED: AVDD3 / AVDD1 bypass capacitor values are not given in the ADAR1000 datasheet -> take from ADAR1000-EVALZ schematic.

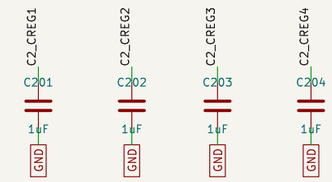
PA_BIASx = 0 V at power-up (DS p41): PA drain stays OFF until firmware sets PA_BIAS OFF = -1.75 V (ADTR1107 VGG_PA abs max -2 V).

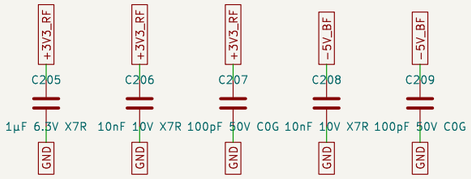
SDO errata (DS p47-49): non-addressed SDO not tristated -> read back one chip at a time (separate CSB).

H2-B GATE-BIAS LIMIT (safety): EXT_PAx_BIAS_OFF/ON and EXT_LNA_BIAS_OFF/ON must stay <= 0x5D (~-1.75 V nominal, V = -code x 4.8/255). NEVER use ADAR1000 Rev B p45 values 0x85 (~-2.5 V) or 0x68 (~-1.96 V): ADTR1107 Rev C abs max VGG_PA -2..0 V, VGG_LNA -2..+0.2 V. Measure PA_BIASx with VDD_PA OFF before enabling the drain.









decoupling per ADI Stingray 02_059522 Rev B (per-tile set); per-pin values not documented

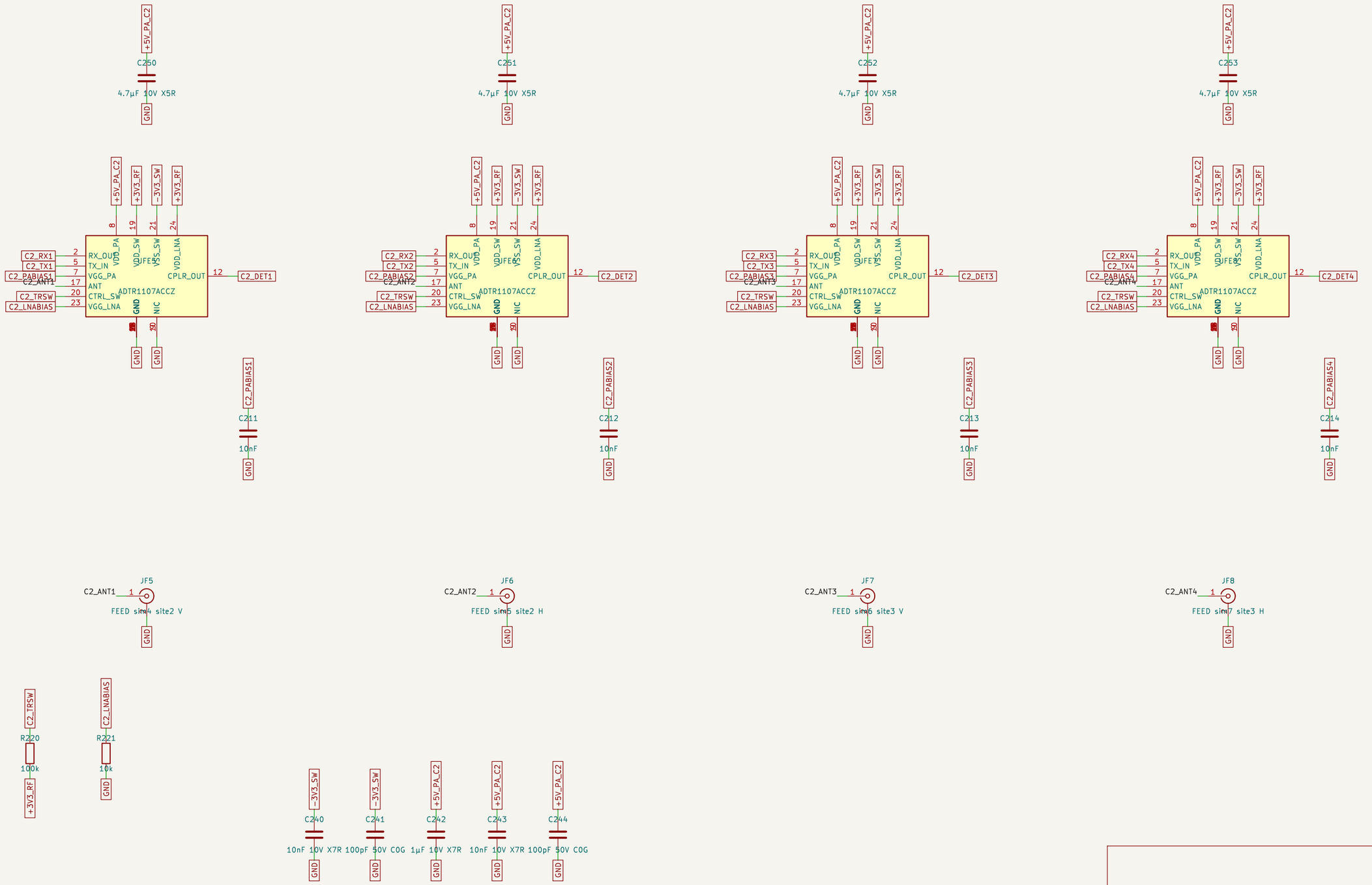
Sheet: /RF_CHAIN_2/BF_2/
File: bf2.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 7/25

UNRESOLVED: VDD_PA / VDD_LNA / VDD_SW / VSS_SW bypass values not in ADTR1107 Rev C (Fig 89 shows only 10 nF on VGG_PA).

Feed ownership = H1_FEED_POLARIZATION_MAP.csv (topology A). LNA kept powered in TX and pinched via LNA_BIAS (Stingray style) — confirm with ADI (ADTR1107 TX sequence lists VDD_LNA = 0 V).



decoupling per ADI Stingray 02_059522 Rev B (per-tile set); per-pin values not documented

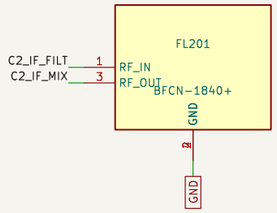
Sheet: /RF_CHAIN_2/FE_2/
File: fe2.kicad_sch

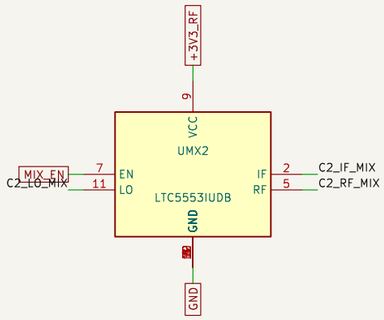
Title:

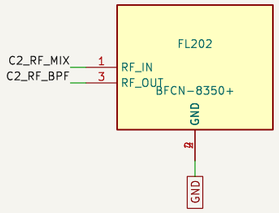
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.9 | Id: 8/25

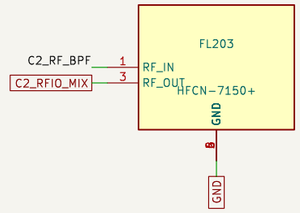
IF 1.75–1.85 GHz: N310 <–> DC block <–> BFCN–1840+ <–> LTC5553 IF (single–ended 50 ohm, no balun).

RF: LTC5553 RF <–> BFCN–8350+ <–> HFCN–7150+ <–> cal coupler (cal sheet) <–> ADAR1000 RF_IO. No DC blocks: RF_IO and LTC5553 RF are both DC–grounded.

















Sheet: /RF_CHAIN_2/CONV_2/
File: conv2.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.9 | Id: 9/25

RF chain 3 (topology A): N310 ch2 IF 1.8 GHz <-> LTC5553 <-> 8.4 GHz <-> cal coupler <-> ADAR1000 UBF3 <-> 4x ADTR1107 <-> 4 feeds.

BF_3

File: bf3.kicad_sch

FE_3

File: fe3.kicad_sch

CONV_3

File: conv3.kicad_sch

J301

C3_IF_N310

1

SMA IF3 (N310 ch2)

GND

SMA footprint (edge-launch vs vertical) UNRESOLVED.

Sheet: /RF_CHAIN_3/
File: chain3.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

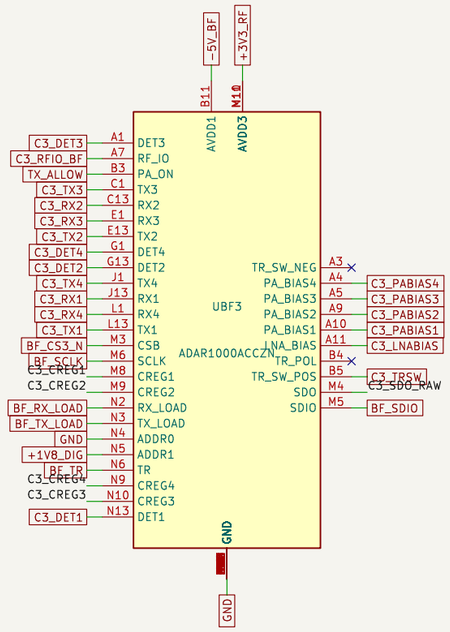
KiCad E.D.A. 9.0.9 | Id: 10/25

UNRESOLVED: AVDD3 / AVDD1 bypass capacitor values are not given in the ADAR1000 datasheet -> take from ADAR1000-EVALZ schematic.

PA_BIASx = 0 V at power-up (DS p41): PA drain stays OFF until firmware sets PA_BIAS OFF = -1.75 V (ADTR1107 VGG_PA abs max -2 V).

SDO errata (DS p47-49): non-addressed SDO not tristated -> read back one chip at a time (separate CSB).

H2-B GATE-BIAS LIMIT (safety): EXT_PAx_BIAS_OFF/ON and EXT_LNA_BIAS_OFF/ON must stay <= 0x5D (~-1.75 V nominal, V = -code x 4.8/255). NEVER use ADAR1000 Rev B p45 values 0x85 (~-2.5 V) or 0x68 (~-1.96 V): ADTR1107 Rev C abs max VGG_PA -2..0 V, VGG_LNA -2..+0.2 V. Measure PA_BIASx with VDD_PA OFF before enabling the drain.



decoupling per ADI Stingray 02_059522 Rev B (per-tile set); per-pin values not documented

Sheet: /RF_CHAIN_3/BF_3/
File: bf3.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.9 | Id: 11/25

UNRESOLVED: VDD_PA / VDD_LNA / VDD_SW / VSS_SW bypass values not in ADTR1107 Rev C (Fig 89 shows only 10 nF on VGG_PA).

Feed ownership = H1_FEED_POLARIZATION_MAP.csv (topology A). LNA kept powered in TX and pinched via LNA_BIAS (Stingray style) — confirm with ADI (ADTR1107 TX sequence lists VDD_LNA = 0 V).

C350 4.7µF 10V X5R

C351 4.7µF 10V X5R

C352 4.7µF 10V X5R

C353 4.7µF 10V X5R

U310 ADTR1107ACCZ

U311 ADTR1107ACCZ

U312 ADTR1107ACCZ

U313 ADTR1107ACCZ

C311 10nF

C312 10nF

C313 10nF

C314 10nF

JF9 FEED sim32 site4 V

JF10 FEED sim33 site4 H

JF11 FEED sim34 site5 V

JF12 FEED sim35 site5 H

R320 100k

R321 10k

C340 10nF 10V X7R

C341 100pF 50V C0G

C342 1µF 10V X7R

C343 10nF 10V X7R

C344 100pF 50V C0G

decoupling per ADI Stingray 02_059522 Rev B (per-tile set); per-pin values not documented

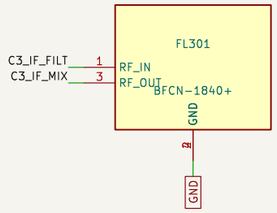
Sheet: /RF_CHAIN_3/FE_3/
File: fe3.kicad_sch

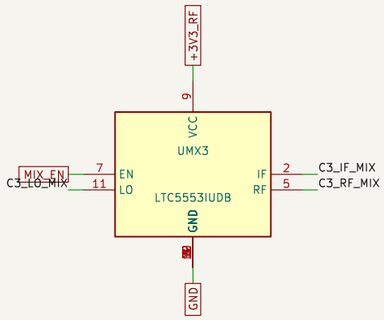
Title:

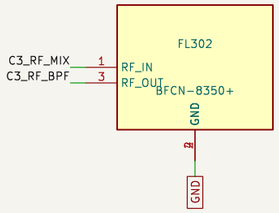
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.9 | Id: 12/25

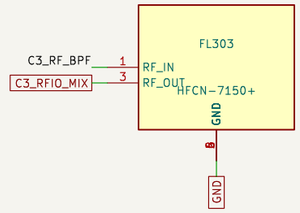
IF 1.75–1.85 GHz: N310 <–> DC block <–> BFCN–1840+ <–> LTC5553 IF (single–ended 50 ohm, no balun).

RF: LTC5553 RF <–> BFCN–8350+ <–> HFCN–7150+ <–> cal coupler (cal sheet) <–> ADAR1000 RF_IO. No DC blocks: RF_IO and LTC5553 RF are both DC–grounded.

















Sheet: /RF_CHAIN_3/CONV_3/
File: conv3.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.9 | Id: 13/25

RF chain 4 (topology A): N310 ch3 IF 1.8 GHz <-> LTC5553 <-> 8.4 GHz <-> cal coupler <-> ADAR1000 UBF4 <-> 4x ADTR1107 <-> 4 feeds.

BF_4

File: bf4.kicad_sch

FE_4

File: fe4.kicad_sch

CONV_4

File: conv4.kicad_sch

J401

C4_IF_N310

1

SMA IF4 (N310 ch3)

GND

SMA footprint (edge-launch vs vertical) UNRESOLVED.

Sheet: /RF_CHAIN_4/
File: chain4.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

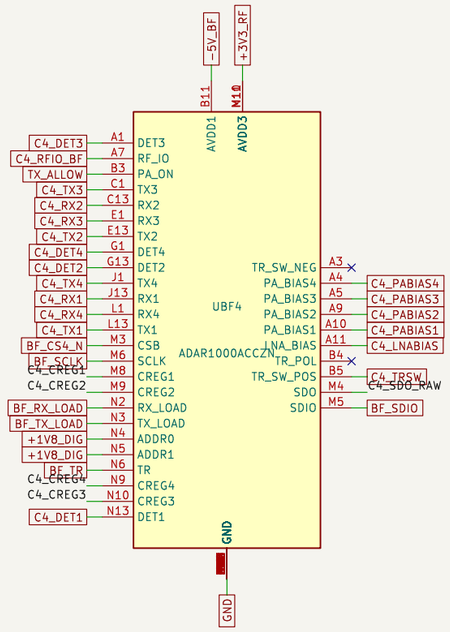
KiCad E.D.A. 9.0.9 | Id: 14/25

UNRESOLVED: AVDD3 / AVDD1 bypass capacitor values are not given in the ADAR1000 datasheet -> take from ADAR1000-EVALZ schematic.

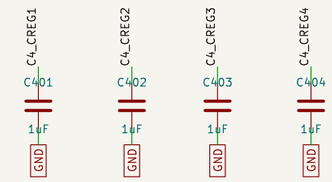
PA_BIASx = 0 V at power-up (DS p41): PA drain stays OFF until firmware sets PA_BIAS OFF = -1.75 V (ADTR1107 VGG_PA abs max -2 V).

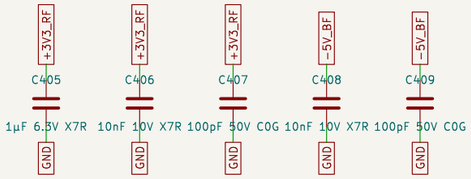
SDO errata (DS p47-49): non-addressed SDO not tristated -> read back one chip at a time (separate CSB).

H2-B GATE-BIAS LIMIT (safety): EXT_PAx_BIAS_OFF/ON and EXT_LNA_BIAS_OFF/ON must stay <= 0x5D (~-1.75 V nominal, V = -code x 4.8/255). NEVER use ADAR1000 Rev B p45 values 0x85 (~-2.5 V) or 0x68 (~-1.96 V): ADTR1107 Rev C abs max VGG_PA -2..0 V, VGG_LNA -2..+0.2 V. Measure PA_BIASx with VDD_PA OFF before enabling the drain.









decoupling per ADI Stingray 02_059522 Rev B (per-tile set); per-pin values not documented

Sheet: /RF_CHAIN_4/BF_4/
File: bf4.kicad_sch

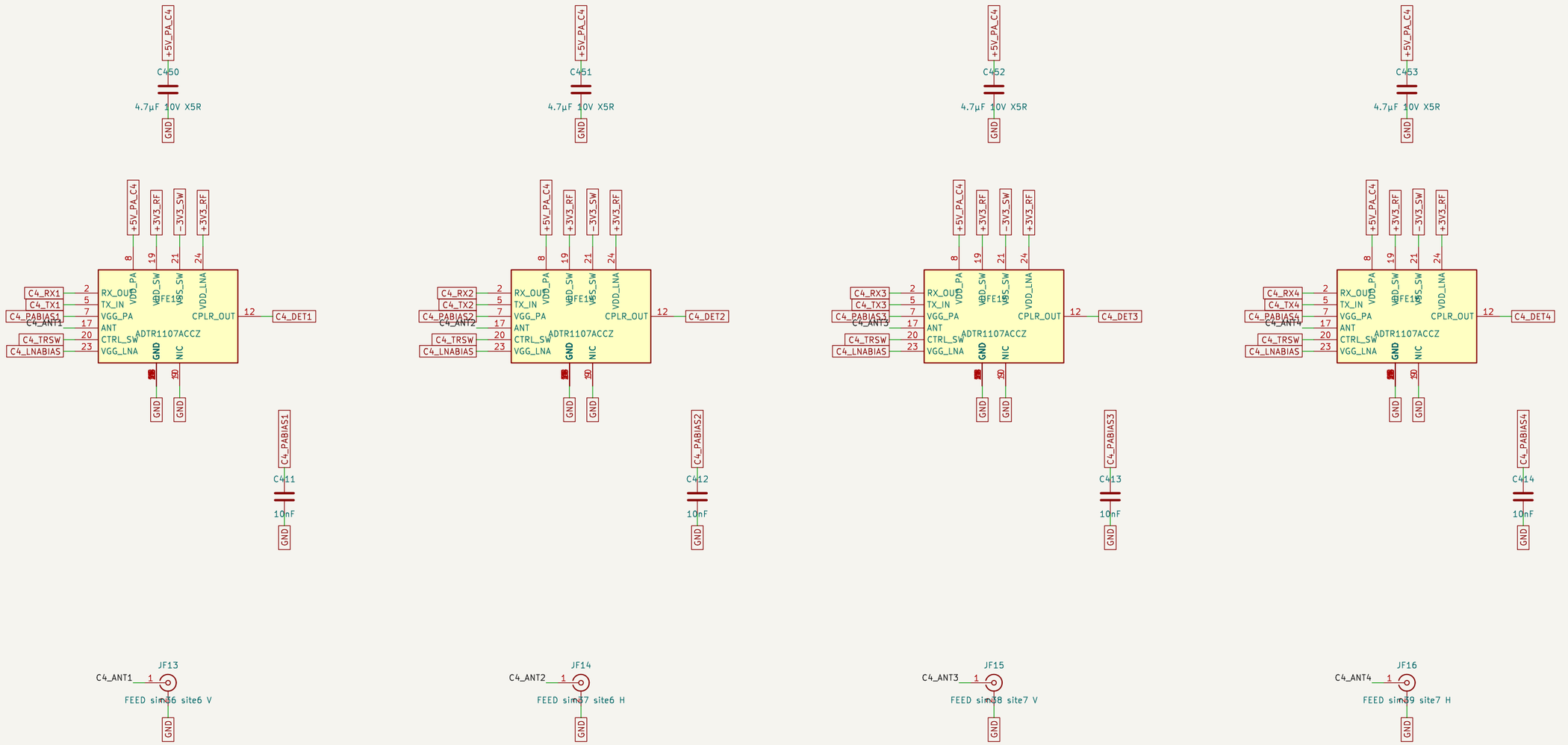
Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.9 | Id: 15/25

UNRESOLVED: VDD_PA / VDD_LNA / VDD_SW / VSS_SW bypass values not in ADTR1107 Rev C (Fig 89 shows only 10 nF on VGG_PA).

Feed ownership = H1_FEED_POLARIZATION_MAP.csv (topology A). LNA kept powered in TX and pinched via LNA_BIAS (Stingray style) — confirm with ADI (ADTR1107 TX sequence lists VDD_LNA = 0 V).



decoupling per ADI Stingray 02_059522 Rev B (per-tile set); per-pin values not documented

Sheet: /RF_CHAIN_4/FE_4/
File: fe4.kicad_sch

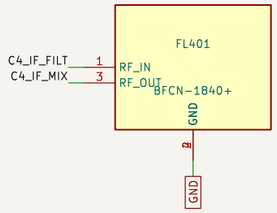
Title:

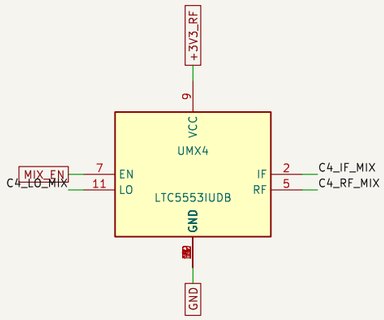
Size: A3 | Date: | **Rev:**

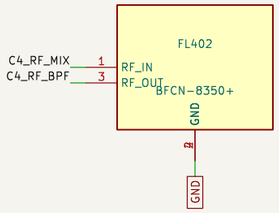
KiCad E.D.A. 9.0.9 | Id: 16/25

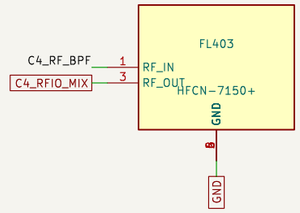
IF 1.75–1.85 GHz: N310 <–> DC block <–> BFCN–1840+ <–> LTC5553 IF (single–ended 50 ohm, no balun).

RF: LTC5553 RF <–> BFCN–8350+ <–> HFCN–7150+ <–> cal coupler (cal sheet) <–> ADAR1000 RF_IO. No DC blocks: RF_IO and LTC5553 RF are both DC–grounded.

















Sheet: /RF_CHAIN_4/CONV_4/
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Size: A3 | Date: | **Rev:**

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UNRESOLVED: loop filter (CPOUT -> VTUNE, RS_SW) needs ADIsimPLL design; pins intentionally left unconnected.

ADF4371 RF8 is ENABLED at power-up by register default (REG0025 RF_EN=1): CE held low by pull-down until firmware programs the part.

LO path: ADF4371 RF8P 6.6 GHz -> AVA-183A+ -> 3 dB pad -> EP4RKU+ 4-way -> LTC5553 LO x4 (C1_LO..C4_LO). Equal-length routing required.

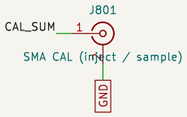
H2-B: loop filter C509/R507/C510/R508/C511 = DERIVED 3rd-order passive (Kv 50 MHz/V nominal, Icp 1.75 mA, N 660, fPFD 10 MHz; 98 kHz BW, 46 deg PM; tools/loop_filter.py) - confirm in ADIsimPLL. RS_SW (fastlock) UNRESOLVED. AVA-183A+ bias L505/C512-C515 per Mini-Circuits Fig 2 (NPO 0402, L1 wire-wound 0805): ML1-ML7 microstrips + pads 1.6 to GND per PL-328.

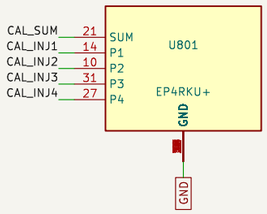
REF_10M
C503
100nF
PLL_REFP
C504
100nF
PLL_REFN
U501
ADF4371BCCZ
+3V3_LO
+5V_LO
GND
PLL_VTUNE
PLL_CS_N
BF_SCLK
PLL_CE
PLL_REFP
PLL_REFN
PLL_CP
PLL_RF8P
PLL_RF8N
PLL_AUXP
PLL_AUXN
BF_SDIO
PLL_MUXOUT_RAW
C501
1uF
C502
1uF
C509
120pF
R507
5.6k
C510
820pF
R508
2.2k
C511
22pF
L501
7.5nH
L502
7.5nH
C505
10pF
LO_AMP_IN
#FLG501
PWR_FLAG
LO_AMP_VD1
C506
10pF
R502
49.9
C512
0.1uF
L505
3.3nH
C513
1nF
C514
18pF
C515
5.6pF
U502
AVA-183A+
VD1
VD2
RF_IN
RF_OUT
EP
NU
LO_AMP_OUT
R504
294
R505
17.4
LO_SPLIT_IN
R506
294
R501
1k
L503
7.5nH
L504
7.5nH
C507
10pF
C508
10pF
R503
49.9
LO_MON
PLL_AUXN_T
U503
EP4RKU+
SUM
P1
P2
P3
P4
C1_LO
C2_LO
C3_LO
C4_LO
J501
SMA LO_MON (RFAUX8)

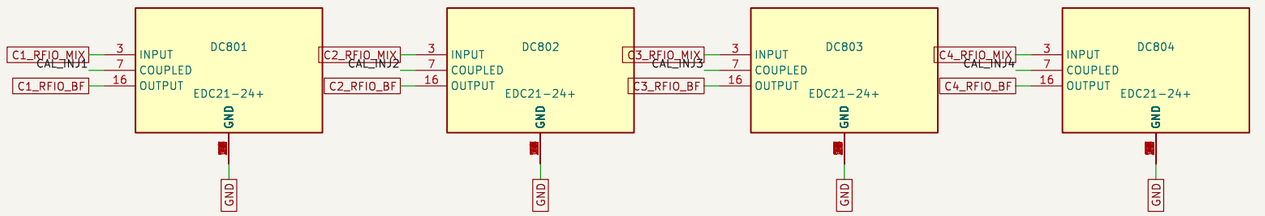
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File: lo.kicad_sch
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KiCad E.D.A. 9.0.9
Id: 18/25

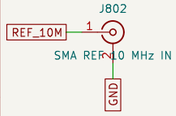
Calibration: N310 relative channel phase is random per start/retune -> run CAL-port RX/TX loopback after every N310 init or LO retune; coupler + EP4RKU+ paths must be phase-matched in layout and characterised once (VNA).

Per-element: ADTR1107 CPLR_OUT -> ADAR1000 DETx (power only, -20..+10 dBm). Element phase cal needs a near-field/OTA probe in a shielded setup (no OTA at 8.4 GHz).

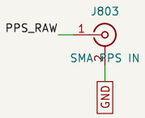


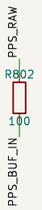




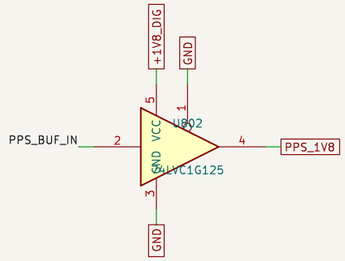












Sheet: /CAL_AND_TEST_IO/		
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KiCad E.D.A. 9.0.9		Id: 19/25

Power-up order (ADAR1000 Rev B p44-45 + ADTR1107 Rev C p24): +1V8_DIG/+3V3_AUX (always on) -> MCU -> +3V3_RF (VDD_SW, VDD_LNA, AVDD3) -> −5V_BF & −3V3_SW (PWR_EN_NEG) -> program PA_BIAS OFF = −1.75 V, CTRL_SW = RX -> +5V_PA rail -> PA_DRAIN_EN (hardware AND) -> raise VGG_PA for 220 mA.

Power-down: TX off (BF_TR low) -> PA_BIAS to −1.75 V -> PA_DRAIN_EN low -> negative rails -> +3V3_RF. A watchdog or NEG_OK fault removes the PA drain in hardware.

All regulator blocks are UNRESOLVED placeholders (TBD_*), not parts: topology, inductors, compensation and feedback values are not designed yet.

#FLG601
PWR_FLAG
VIN_12V
#FLG602
PWR_FLAG
GND
J601
VIN_12V
GND
VIN 12V bench

H2-B PA drain switches U621-U624 (TI TPS22810 SLVSDH0C): QOD tied to VOUT (internal RPD 250-400 ohm at 5 V -> fast drain discharge on disable); CT = 2200 pF (TI test condition; tR ~0.5-0.7 ms); CIN = 1 uF per switch (TI recommended). EN/UVLO VENR 1.13-1.30 V < 1.8 V logic high.

#FLG603
PWR_FLAG
+3V3_RF_SRC
R602
10m
+3V3_RF

#FLG611
PWR_FLAG
+5V_PA_IN_C1
R611
10m
+5V_PA_IN_C1
C625
1uF
GND
#FLG612
PWR_FLAG
+5V_PA_IN_C2
R612
10m
+5V_PA_IN_C2
C626
1uF
GND
#FLG613
PWR_FLAG
+5V_PA_IN_C3
R613
10m
+5V_PA_IN_C3
C627
1uF
GND
#FLG614
PWR_FLAG
+5V_PA_IN_C4
R614
10m
+5V_PA_IN_C4
C628
1uF
GND
+5V_PA_IN_C1
PA_DRAIN_EN
U621
TPS22810DRV
VIN
EN/UVLO
VOUT
QOD
CT
GND
+5V_PA_C1
+5V_PA_C1
PA_SW1_CT
U622
TPS22810DRV
+5V_PA_IN_C2
+5V_PA_C2
PA_SW2_CT
U623
TPS22810DRV
+5V_PA_IN_C3
+5V_PA_C3
PA_SW3_CT
U624
TPS22810DRV
+5V_PA_IN_C4
+5V_PA_C4
PA_SW4_CT
PA_SW1_CT
C621
2200pF
GND
PA_SW2_CT
C622
2200pF
GND
PA_SW3_CT
C623
2200pF
GND
PA_SW4_CT
C624
2200pF
GND

REC drain crowbar per chain c (H2C_SEQUENCING_SIM.md rev 2, items 1-4):
Q96c CSD16301Q2 (SLPS235E): drain -> PWR_CB_Rc -> 3x 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065) -> +5V_PA_Cc (U62c VOUT);
gate PWR_CB_Gc pulled up 4.7k from +5V_PA_IN_Cc (U62c VIN side); Q97c DMG1012T (Vth 0.5-1.0 V, Rds 0.7R @ 1.8 V) holds gate low
while PA_DRAIN_EN is high via 10k / 1nF COG (PWR_CB_Dc). Break-before-make on fault, make-before-break on arm.

+5V_PA_IN_C1
R902 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065)
R900 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065)
R901 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065) pulse
+5V_PA_C1
PWR_CB_G1
PWR_CB_R1
PWR_CB_D1
Q961 CSD16301Q2
Q971 DMG1012T
R906 10R 1% 1nF 50V C0G
PA_DRAIN_EN
PWR_CB_D1
GND
PWR_CB_D1
+5V_PA_IN_C2
R903 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065)
R904 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065)
R905 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065) pulse
+5V_PA_C2
PWR_CB_G2
PWR_CB_R2
PWR_CB_D2
Q962 CSD16301Q2
Q972 DMG1012T
R907 10R 1% 1nF 50V C0G
PA_DRAIN_EN
PWR_CB_D2
GND
PWR_CB_D2
+5V_PA_IN_C3
R906 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065)
R907 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065)
R908 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065) pulse
+5V_PA_C3
PWR_CB_G3
PWR_CB_R3
PWR_CB_D3
Q963 CSD16301Q2
Q973 DMG1012T
R908 10R 1% 1nF 50V C0G
PA_DRAIN_EN
PWR_CB_D3
GND
PWR_CB_D3
+5V_PA_IN_C4
R909 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065)
R910 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065)
R911 4.7R 1% 1R RCS1206 anti-surge in parallel (0.333 R; Vishay 20065) pulse
+5V_PA_C4
PWR_CB_G4
PWR_CB_R4
PWR_CB_D4
Q964 CSD16301Q2
Q974 DMG1012T
C934 50V C0G
PA_DRAIN_EN
PWR_CB_D4
GND
PWR_CB_D4

+3V3_RF
+3V3_RF_SRC
U610 INA226 Vbus VS A1 A0 SDA SCL Vin+ Vin− GND Alert
+5V_PA_IN_C1
I2C3_SDA
I2C3_SCL
+5V_PA
+5V_PA_C1
INA_ALERT_N
U611 INA226
+5V_PA_IN_C2
+5V_PA_C2
U612 INA226
+5V_PA_IN_C3
+3V3_AUX
+5V_PA_C3
U613 INA226
+5V_PA_IN_C4
+5V_PA_C4
U614 INA226
+3V3_AUX

#FLG620
PWR_FLAG
PCA_VREF2
R603 200k
I2C3_SCL
R604 4.7k
I2C3_SDA
R605 4.7k
INA_ALERT_N
R606 10k
+1V8_DIG
+3V3_AUX
U615 PCA9306 VREF1 VREF2 EN SCL1 SCL2 SDA1 SDA2 GND
I2C_SCL
I2C_SDA
C629 100pF 50V C0G
GND

H2-C: U610 A0 -> SDA (A1 = GND) = 0x42 (INA226 SBOS547C Table 6-2). was GND = 0x40 duplicate of U611.
C629 100 pF on PCA_VREF2/EN (PCA9306 SCPS1130 9.2.2.1).

Sheet: /POWER_SEQUENCING/
File: power.kicad_sch
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Size: A2 | Date: | Rev:
KiCad E.D.A. 9.0.9 | Id: 20/25

Safety chain: reset/power-up -> all enables pulled low -> TX muted, PA drain off. Watchdog timeout -> WDT_OK low -> TX_ALLOW and PA_DRAIN_EN drop in hardware, independent of firmware.

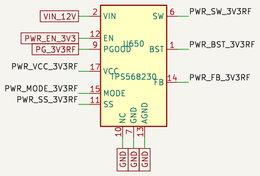
Beam updates: BF_TX_LOAD / BF_RX_LOAD are shared by all four ADAR1000 (synchronous latch); never pulse both within 8 SCLK (DS p.33-34).

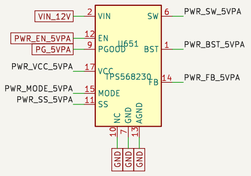
REC (H2C_SEQUENCING_SIM rev 2 item 7): R735 1k in series U705 Y -> PA_DRAIN_EN (BAT54C D961 clamp to PG_3V3RF on PA_INTERLOCK sheet).

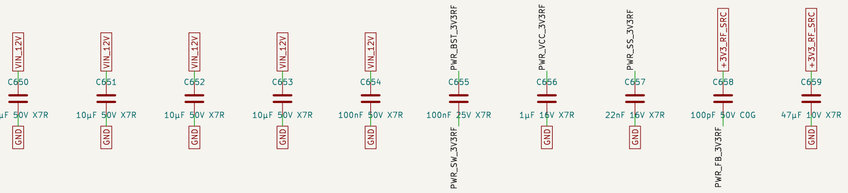
H2-B watchdog U702 (TPS3430 SBVS366A Table 7-6): CWD = 10k to VDD, SETO = SET1 = GND -> window tWDL 9.0 ms (max 10.35) / tWDU 109 ms (min 92.7): kick WDI every 20-80 ms. CRST = 10k to VDD -> tRST 10 ms. SET pins hard-strapped so firmware cannot select 'watchdog disabled' (SETO=1, SET1=0). C702 = VDD1 0.1 uF bypass (DS recommended).

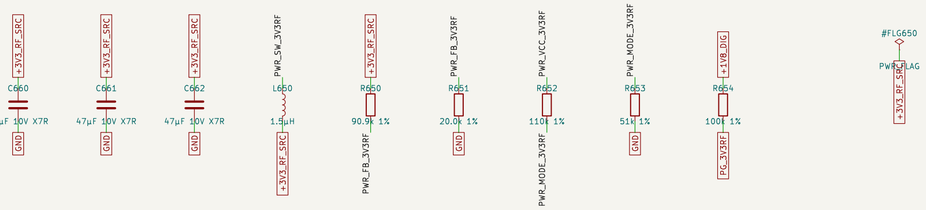
U701 decoupling per ST DS12589 Fig 16: VDD pins 23/35/48 = C703/C704/C705 100 nF + C706 4.7 uF; VDDA pin 21 = C707 10 nF + C708 1 uF; VREF+ pin 20 = C709 100 nF + C710 1 uF (all on +1V8_DIG). VREF+ is tied to VDDA: ST requires VREF+ = VDDA when VDDA < 2 V (DS12589 3.11.1). VBAT pin 1: no capacitor (no value in DS12589) - UNRESOLVED. PC4 (pin 16) = PG_5VPA input (U651 PGOOD, 100k pull-up R676 on PWR_MAIN).

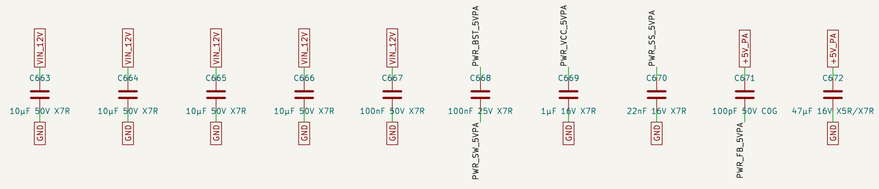
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Size:	A2
Date:	
Rev:	
KiCad E.D.A.	9.0.9
Id:	21/25

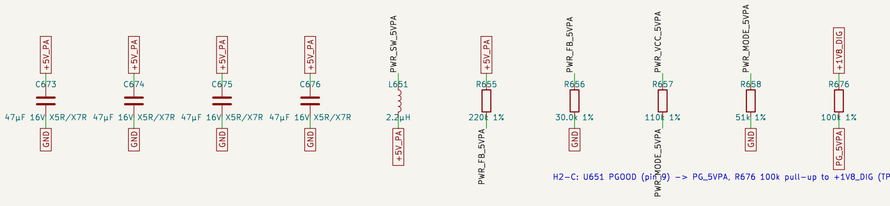




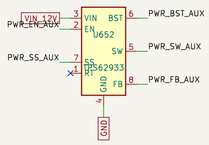


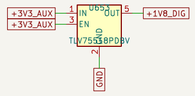


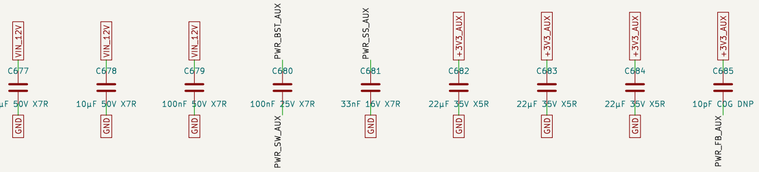


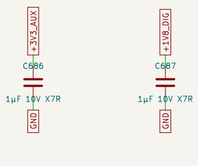


H2-C: U651 PGOOD (pin 9) -> PG_5VPA, R676 100k pull-up to +1V8_DIG (TPS568230 SLVSEY5C 7.3.4). to MCU U701 PC4 (pin 16).

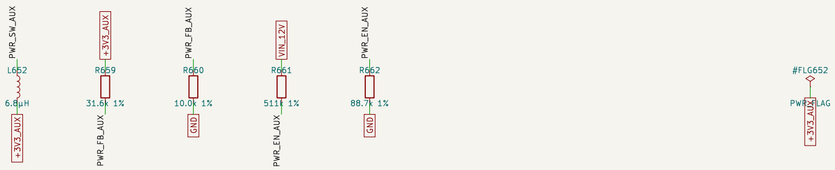




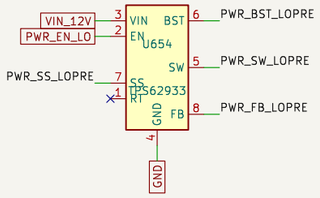


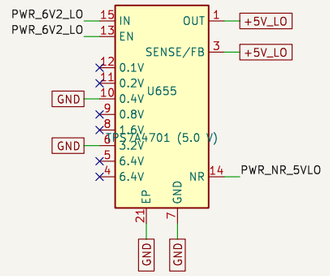


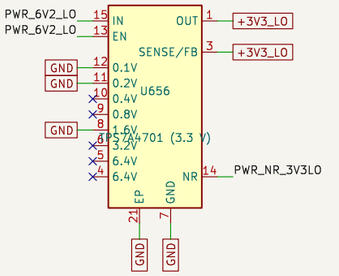
PWR_MAIN (H2-C POWER_SPEC): U650/U651 TPS568230 per TI SLVSEY5C (Aug 2019); U652 TPS62933 per TI SLUSEA4D (Aug 2022); U653 TLV75518P per TI SBVS320D (Sep 2024). U651 PGOOD left open (unused). U652 RT floating = 500 kHz (SLUSEA4D Table 7-1). L650/L651 MPN+footprint UNRESOLVED. C685 DNP.

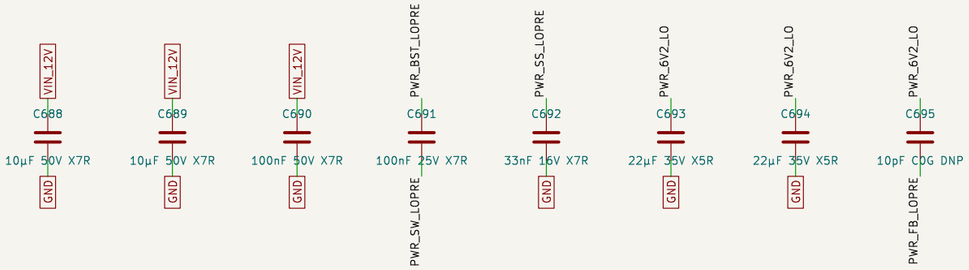


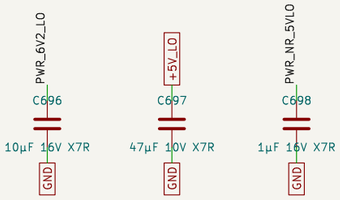
Sheet: /PWR_MAIN/	
File: pwr_main.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 9.0.9	**Rev:** Id: 22/25

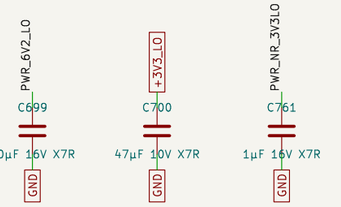


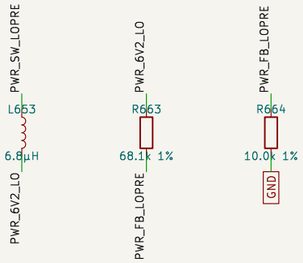








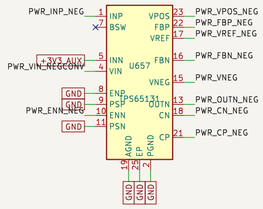


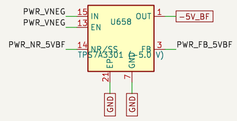


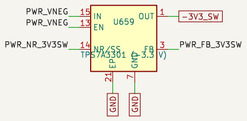


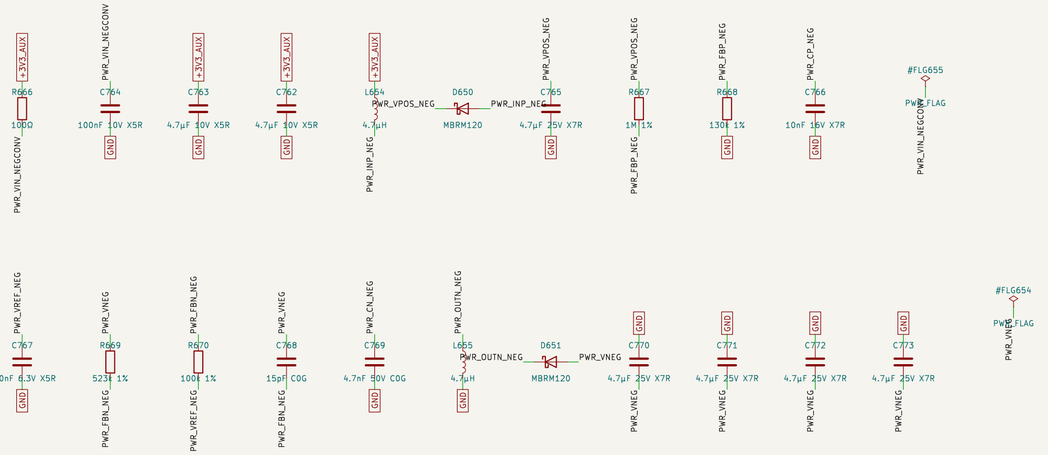
, unused program pins left floating ("If not used, leave this pin floating", pin table). EN of U655/U656 tied to IN. R665 omitted (pull-down on MCU sheet, AMENDMENTS A3). C701 renumbered C761 (ref clash with mcu.kicad_sch).

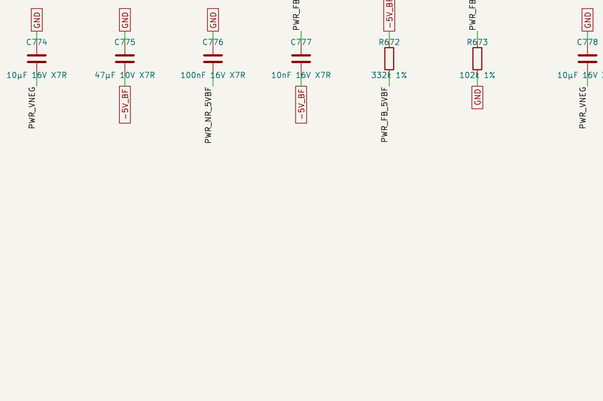
Sheet: /PWR_LO/	
File: pwr_lo.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 9.0.9	Id: 23/25

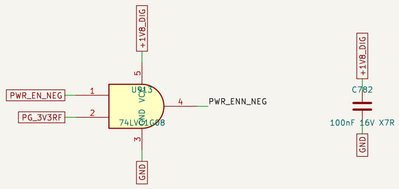






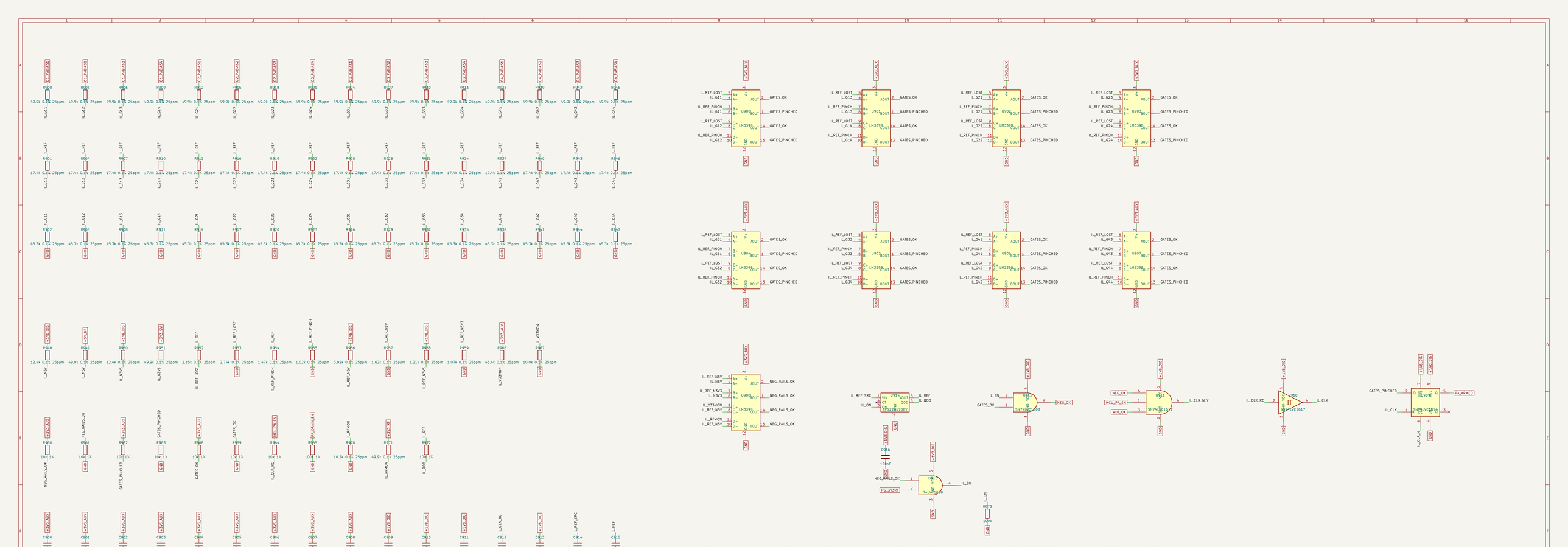






che BSW pin floating"); U658/U659 TPS7A3301 per TI SBVS169D (Apr 2015), EN tied to IN. U913 74LVC1G08 (TI SN74LVC1G08): ENN = PWR_EN_NEG AND PG_3V3RF (A2). R671 omitted (A3). C702–C722 renumbered C762–C782 (ref clash with mcu.kicad_sch / A4).

Sheet: /PWR_NEG/		
File: pwr_neg.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 24/25


TY INTERLOCK (H2-C. INTERLOCK_SPEC + AMENDMENTS A1/A3/A6)
holds (ratiometric to +1V8_DIG / IL_REF, 0.1% resistors):
1.009 V -> PA gate = −0.15 V (worst −0.19…−0.11 V) pulls GATES_OK low
H 0.737 V -> gate < −1.50 V (worst −1.58…−1.42 V) releases GATES_PINCHED
V (worst 2.94…3.42 V); −3V3_SW trip −3.20 V (worst −3.13…−2.87 V)
ADAR1000 bias (0.5 V) is independent of this network.
D GATES_OK (U913), IL_CLR_N, PG_PA_EN AND WDT_OK AND NEG_OK (U911).
MCU_PA_EN (delayed 5.5–8.9 us (R984/C912 + U910 Schmitt)). 0 = GATES_PINCHED.
ED AND WDT_OK AND NEG_OK (U705 on MCU sheet). R965 100k pull-down.
FAIL-SAFE BEHAVIOUR
OK, GATES_OK, GATES_PINCHED: 10k to +3V3_AUX / 10k to GND (high 1.57–1.73 V).
comparators unpowered, nets pulled to 0 V -> NEG_OK low -> drain off.
gate-reference supply) = +1V8_DIG via U914 TPS22917, ON = NEG_RAILS_OK.
ff and discharged (VDD via R972 100R): no positive bias into unpowered PA gates.
n sit below GND (<=−0.1 mA via 49.9k, LM339B abs max −50 mA); their outputs are
6.0.2) but NEG_OK is already forced low by NEG_RAILS_OK through U912.
D low via IL_CLR_N; re-arm only by MCU_PA_EN low->high with gates pinched.
absent -> logic unpowered (off), R965 holds PA_DRAIN_EN low.
5 9.1.2, VDL full VD (IN− 10k to 9+, IN+ 10k to GND), output left open (9.1.3).
F: TPS22917 SLVSDW8B; SN74LVC1G74 SCES394G; LVC1G17 SCES351Y; LVC1G11 SCES487I.
REC hardware (H2C_SEQUENCING_SIM.md rev 2, items 5–9; parts: REC_PARTS.md / IMPL_REC.md):
– U920 TPS386000 (SBVS105F): SENSE1 +1V8_DIG 1.671 V (R977 32.4k / R978 10.2k), SENSE2 +3V3_AUX 2.969 V (R979 64.2k / R1000 10.0k);
RESET1/RESET2 open-drain wired-AND onto PG_3V3RF: tD = 9.6 ms (C71/C72 2.2 nF, Eq B); outputs valid for VDD >= 0.9 V.
VDD = IL_SUP_VDD = BAT54C D960 diode-OR of +3V3_RF / +5V_PA, C935 100 nF, MR = VDD, WDI = GND, SENSE3/4L/4H = GND,
unused open-drain RESET3/RESET4/WDO joined on IL_SUP_DD with R1001 1M pull-up; CT3, CT4 open (fixed delay, ps). VREF open (DD0 test cond.).
– Active-high 2nd opinion: RESET2 (IL_SUP_DD) -> Q995 (DMN63D8LDW) -> IL_LCK_RC -> Q965 (D3)
clamps TPS22917 ON (IL_CK1) R974 10k from IL_EN to IL_ON, U912 A stage = IL_LEN.
– D961 BAT54C: cathode PG_3V3RF, anodes PA_DRAIN_EN and IL_CLR_N: tk series R735 (U705 Y, mcu sheet) and R975 (U911 Y).
– U921 TLV755B8P: IN +3V3_RF (C938 1 uF), EN = PG_3V3RF, OUT IL_REF_SRC (C914 1 uF) -> U914 VIN (was +1V8_DIG).
– C915 IL_REF decoupling 100 nF -> 10 nF
– U908 ch D: +3V3_RF UV 3.10 V: R971 49.9k / R970 10.2k (IL_RFMON) on pin 11; pin 10 = IL_REF_NSV, pin 13 -> NEG_RAILS_OK.
– Crowbars (items 1–4) are on the POWER_SEQUENCING sheet next to U821–U824.
OPEN: supervisor sink current with clamps −2.4 mA > 1 mA VOL spec; TLV755 EN VLO 0.3 V not guaranteed (see IMPL_REC.md).
Sheet: /PA_INTERLOCK/
File: interlock.kicad_sch
Title:
Size: A1
Date:
Rev:
KiCad E.D.A. 9.0.9
Id: 25/25